Chapter 4: Functions of Several Variables

Calculus II

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Section 1

Functions of several variables

Function $\mathbf{F}$: Definition

We use the notation

$$\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$$

for a function $\mathbf{F}$ that assigns a vector $\mathbf{F}(\mathbf{X})$ in $\mathbb{R}^m$ to each vector $\mathbf{X}$ in a subset D of $\mathbb{R}^n$, and say that $\mathbf{F}$ is a function from D to $\mathbb{R}^m$.

- When the domain D of a function $\mathbf{F}$ of n variables is not specified we assume, as we have with functions of a single variable, that the domain is the largest set for which the definition makes sense.

- We call the set of outputs $\mathbf{F}(D)$ the following:

$$\mathbf{F}(D) = \begin{cases} \text{the range of } \mathbf{F} \\ \text{or} \\ \text{the image of } D \end{cases}$$

Function $\mathbf{F}$: Properties and Notation

(a) Normally, We say $\mathbf{F}$:

- One to One: if $\mathbf{F}(\mathbf{U}) = \mathbf{F}(\mathbf{V})$ only when $\mathbf{U} = \mathbf{V}$.
- Maps D onto a set B in $\mathbb{R}^m$: if $\mathbf{F}(D) = B$.

(b) We usually denote:

- A function whose output is a vector by a bold capital letter $\mathbf{F}$.
- A function whose output is a number is called a scalar valued, or real valued, function and we usually denote it by a lower case letter f.

(c) If f and $\mathbf{F}$ are functions, then $f(\mathbf{X})$ and $\mathbf{F}(\mathbf{X})$ are values assigned to $\mathbf{X}$.

Definition 4.1.1

A function $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$ assigns a vector $\mathbf{F}(\mathbf{X})$ in $\mathbb{R}^m$ to each $\mathbf{X}$ in D, denoted

$$\mathbf{F}(\mathbf{X}) = (f_1(\mathbf{X}), f_2(\mathbf{X}), \ldots, f_m(\mathbf{X}))$$

The function f_j is called the j-th component function of $\mathbf{F}$.

Constant function

Definition 4.1.2

A function that assigns the same vector $\mathbf{C}$ in $\mathbb{R}^m$ to each vector $\mathbf{X}$ in $\mathbb{R}^n$ is called a **constant function**.

Example 4.1.3

The functions $f(x, y) = 7$ and $\mathbf{G}(x, y) = (8, 3, 2)$ are examples of constant functions. f is a constant function from $\mathbb{R}^2$ to $\mathbb{R}$, and $\mathbf{G}$ is a constant function from $\mathbb{R}^2$ to $\mathbb{R}^3$.

Linear function

Definition 4.1.4

A function $\mathbf{L}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ is said to be linear if

$$a\mathbf{L}(\mathbf{U}) = \mathbf{L}(a\mathbf{U}), \text{ and } \mathbf{L}(\mathbf{U}+\mathbf{V}) = \mathbf{L}(\mathbf{U}) + \mathbf{L}(\mathbf{V}) \tag{4.1}$$

for all numbers a and all vectors $\mathbf{U}$ and $\mathbf{V}$ in $\mathbb{R}^n$.

Example 4.1.5

Is $\mathbf{L}(x, y) = (2x - 3y, x, 5y)$ a linear function from $\mathbb{R}^2$ to $\mathbb{R}^3$? Let's check to see whether (4.1) holds.

Let a be a number, and let (x, y) and (u, v) be vectors in $\mathbb{R}^2$.

$$a\mathbf{L}(x, y) = a(2x - 3y, x, 5y) = (2ax - 3ay, ax, 5ay) = \mathbf{L}(ax, ay) = \mathbf{L}(a(x, y))$$

Also,

$$\mathbf{L}(x, y) + \mathbf{L}(u, v) = (2x - 3y, x, 5y) + (2u - 3v, u, 5v)$$
$$= (2(x + u) - 3(y + v), x + u, 5(y + v)) = \mathbf{L}(x + u, y + v).$$

So $\mathbf{L}$ is linear.

Question:
Show that a constant function $\mathbf{F}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ is linear if and only if

$$\mathbf{F}(x_1, x_2, \ldots, x_n) = \mathbf{0}.$$

Recall:
According to linear algebra, every linear function ℓ from $\mathbb{R}^n$ to $\mathbb{R}$ is of the form

$$\ell(x_1, x_2, \ldots, x_n) = c_1x_1 + c_2x_2 + c_3x_3 + \ldots + c_nx_n$$

which we can write as

$$\ell(\mathbf{X}) = \mathbf{C} \cdot \mathbf{X}$$

for some $\mathbf{C} = (c_1, c_2, \ldots, c_n)$ in $\mathbb{R}^n$.

Let's look again at why a linear function $\mathbf{L}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be represented using matrix multiplication.

- **Component Functions**:
  - Let ℓ_k be the k-th component function of $\mathbf{L}$, where $k = 1, 2, \ldots, m$.
  - Since $\mathbf{L}$ is linear, each component $\ell_k(\mathbf{X})$ is also linear.
- **For each** $k = 1, 2, \ldots, m$:
  - There exists a vector $\mathbf{C}_k \in \mathbb{R}^n$ such that:$\ell_k(\mathbf{X}) = \mathbf{C}_k \cdot \mathbf{X}$.
- **Matrix Definition**:
  - Let $\mathbf{C}_k = (c_{k1}, c_{k2}, \ldots, c_{kn})$ be row vectors.
  - Define the matrix $\mathbf{C}$ whose k-th row is $\mathbf{C}_k$.

  The relations $\ell_k(\mathbf{X}) = \mathbf{C}_k \cdot \mathbf{X}$ can be expressed as:

$$\mathbf{L}(\mathbf{X}) = \mathbf{C}\mathbf{X}.$$

The linear function $\mathbf{L}$ can be represented as $\mathbf{L}(\mathbf{X}) = \mathbf{C}\mathbf{X}$.

Theorem 4.1.6

Every linear function $\mathbf{L}$ *from* $\mathbb{R}^n$ *to* $\mathbb{R}^m$ *can be written in matrix form*

$$\mathbf{L}(\mathbf{X}) = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{CX} \tag{4.2}$$

$\mathbf{C}$ *is called the matrix of* $\mathbf{L}$ *or the matrix that represents* $\mathbf{L}$.

Quantization of matrix size

Definition 4.1.7

For an $m \times n$ matrix $\mathbf{C} = [c_{ij}]$, the norm $\|\mathbf{C}\|$ of $\mathbf{C}$ is defined by

$$\|\mathbf{C}\| = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij}^2} \tag{4.3}$$

Important relation

There is an important relation between the norms of $\mathbf{X}, \mathbf{CX}$, and $\mathbf{C}$.

Theorem 4.1.8

Let $\mathbf{C}$ be an $m \times n$ matrix. Then

$$\|\mathbf{CX}\| \leq \|\mathbf{C}\|\|\mathbf{X}\|$$

for every vector $\mathbf{X}$ in $\mathbb{R}^n$.

$C_k = (c_{k1}, c_{k2}, \cdots c_{kn})$

Proof. The k-th component of $\mathbf{CX}$ is the dot product of the k-th row of $\mathbf{C}$,

$$\mathbf{C}_k = (c_{k1}, c_{k2}, \ldots, c_{kn})$$

with $\mathbf{X} = (x_1, x_2, \ldots, x_n)$. By the definition of norm,

$$\|\mathbf{CX}\| = \sqrt{(\mathbf{C}_1 \cdot \mathbf{X})^2 + (\mathbf{C}_2 \cdot \mathbf{X})^2 + \cdots + (\mathbf{C}_m \cdot \mathbf{X})^2}$$

$\leq |C_1 \cdot X| + \cdots + |C_m X|$

Continue

According to the Cauchy-Schwarz inequality, applied to each component this is

$$\leq \sqrt{\|\mathbf{C}_1\|^2 \|\mathbf{X}\|^2 + \|\mathbf{C}_2\|^2 \|\mathbf{X}\|^2 + \cdots + \|\mathbf{C}_m\|^2 \|\mathbf{X}\|^2}$$

$$= \|\mathbf{X}\| \sqrt{\|\mathbf{C}_1\|^2 + \|\mathbf{C}_2\|^2 + \cdots + \|\mathbf{C}_m\|^2} = \|\mathbf{X}\| \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij}^2} = \|\mathbf{C}\| \|\mathbf{X}\|$$

The theorem is proved.

Remark: Combining **Theorems 4.1.6** and **4.1.8** we see that for every linear function $\mathbf{L}$ from $\mathbb{R}^n$ to $\mathbb{R}^m$

$$\|\mathbf{LX}\| \leq \|\mathbf{C}\| \|\mathbf{X}\|.$$

That is, the norm of the output of a linear function is **not greater than** $\|\mathbf{C}\|$ times the norm of the input.

Functions from $\mathbb{R}^n$ to $\mathbb{R}$

Next we look at some examples of functions from $\mathbb{R}^n$ to $\mathbb{R}$.

Example 4.1.9

The area a of a rectangle that has length x, width y is given by $a(x, y) = xy$.

The domain of a is the set D of all ordered pairs (x, y) with $x > 0$ and $y > 0$,

$$a : D \subset \mathbb{R}^2 \to \mathbb{R}.$$

Notice that even though the rule for a makes sense for all ordered pairs (x, y), we restricted the domain based on the context of the problem.

Example 4.1.10

The volume v of a rectangular box that has length x, width y, and height z is given by the function $v(x, y, z) = xyz$.

The domain D of v, is the set of all ordered triples (x, y, z) with $x > 0, y > 0$, and $z > 0$.

The function $v : D \subset \mathbb{R}^3 \to \mathbb{R}$.

Example 4.1.11

Let $g(x, y)$ be defined by:

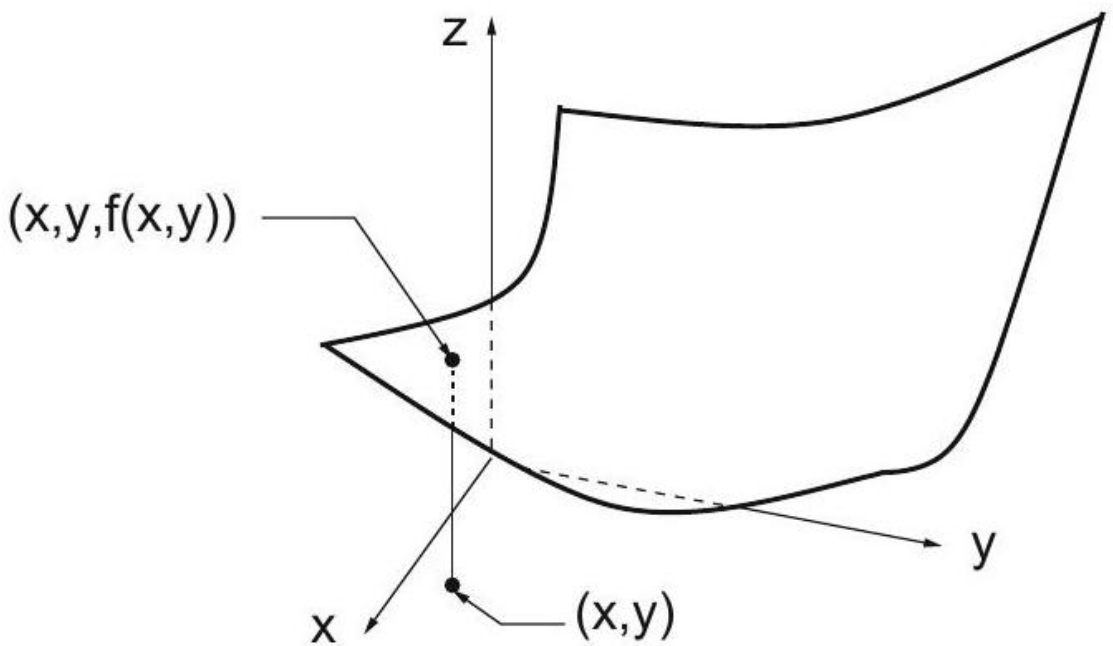
$$g(x, y) = \sqrt{x^2 + y^2}, \quad g : \mathbb{R}^2 \to \mathbb{R}.$$

$g(x, y)$ is the norm of (x, y), namely, $g(x, y) = \|(x, y)\|$.

Graphs of functions from $\mathbb{R}^2$ to $\mathbb{R}$

In single variable calculus we visualized a function f by sketching its graph, the set of all points $(x, f(x))$.

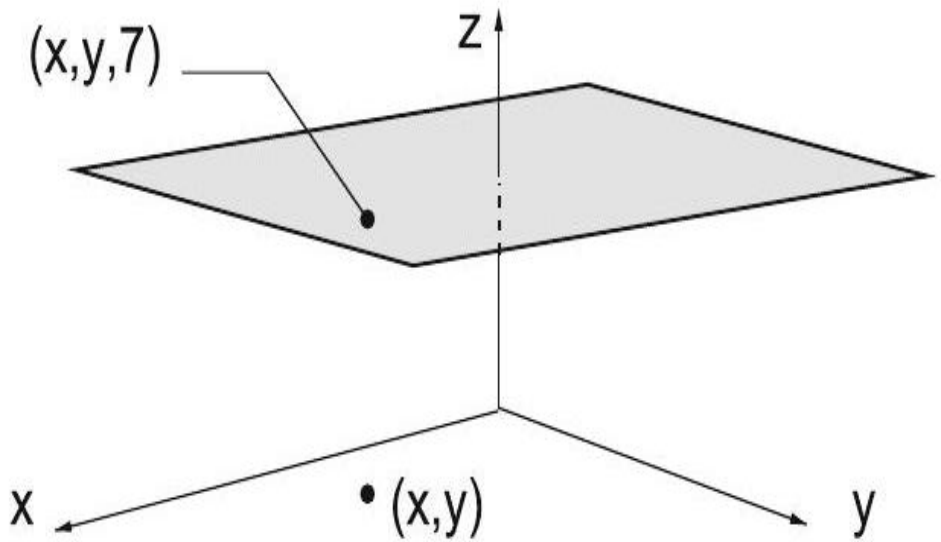
For a function from $\mathbb{R}^2$ to $\mathbb{R}$, the graph of f is the set of ordered triples $(x, y, f(x, y))$, and often looks like a surface in (x, y, z).



Next we sketch the graph of a few functions from $\mathbb{R}^2$ to $\mathbb{R}$.

Example 4.1.12

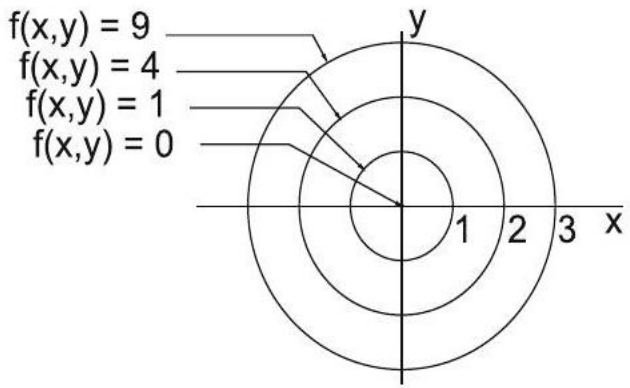
The following figure shows the graph of the constant function $f(x, y) = 7$.

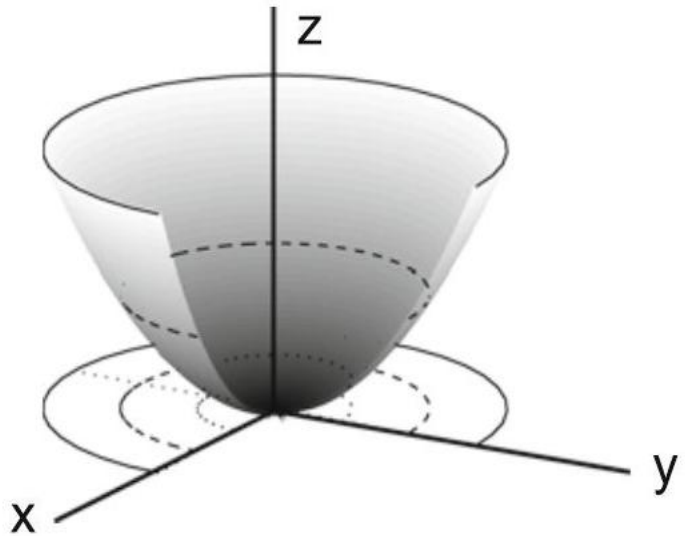


Example 4.1.13

To sketch the graph of $f(x, y) = x^2 + y^2$, we notice that the points in the domain where $f(x, y) = c, c \geq 0$, lie on a circle of radius $\sqrt{c}$. The circle of radius zero is one point.

- On the left side of the following figure, we sketched the points in the domain satisfying $x^2 + y^2 = c$ for $c = 0, 1, 4, 9$.
- On the right hand, we sketched the corresponding points on the graph of f.





Level set

Definition 4.1.14

Let $f : D \subset \mathbb{R}^n \to \mathbb{R}$ be a real valued function, and c a number.

The set of all points $(x_1, x_2, \ldots, x_n)$ in the domain D where

$$f(x_1, x_2, \ldots, x_n) = c$$

is called the c level set of f.

- For functions of two variables, we can draw level sets in the domain and plot corresponding points (x, y, c) on the graph, as in **Example 4.1.13**.
- The set of points on the graph that correspond to a level set $f(x, y) = c$ is called the **contour curve** of the graph at $z = c$.

Example 4.1.15

Question: Let $f(x,y) = \frac{x^2-y^2}{x^2+y^2}$, $(x,y) \neq (0,0)$. Find the level set $c = \frac{x^2-y^2}{x^2+y^2}$, or equivalently:

$$x^2 - y^2 = c(x^2 + y^2), \quad x^2 + y^2 \neq 0.$$

- **Case 1:** $c = 1$
  - The equation becomes $x^2 - y^2 = x^2 + y^2$.
  - Simplifying, we get $y = 0$.
  - The level set for $c = 1$ is the x-axis (excluding the origin).
- **Case 2:** $c = -1$
  - The equation becomes $x^2 - y^2 = -(x^2 + y^2)$.
  - Simplifying, we get $x = 0$.
  - The level set for $c = -1$ is the y-axis (excluding the origin).

Example 4.1.16 (Continued)

- **General Case:** $c \neq -1$
  - Divide by x^2 (assuming $x \neq 0$):

$$1 - \frac{y^2}{x^2} = c\left(1 + \frac{y^2}{x^2}\right).$$

  - Solve for $\left(\frac{y}{x}\right)^2$:

$$\left(\frac{y}{x}\right)^2 = \frac{1-c}{1+c}.$$

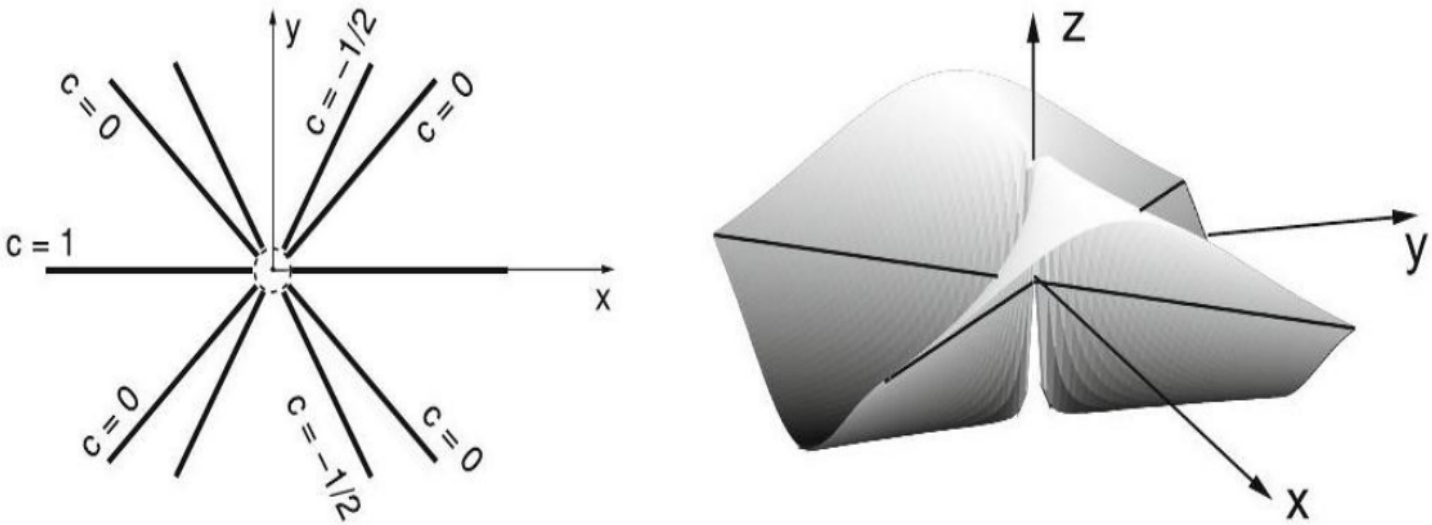
  - The level set consists of two lines:

$$y = \sqrt{\frac{1-c}{1+c}}\,x \quad \text{and} \quad y = -\sqrt{\frac{1-c}{1+c}}\,x.$$

Continue

So for example, the level set for $c = -\frac{1}{2}$ is the pair of lines $y = \sqrt{3}x, y = -\sqrt{3}x$, origin deleted.

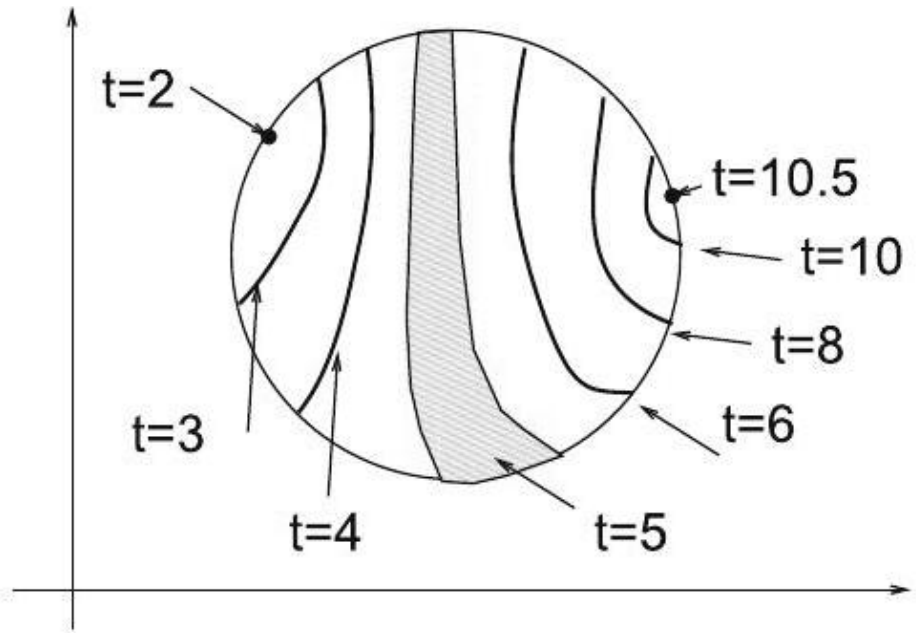
In the following figure, we sketched the level sets for $c = -\frac{1}{2}, 0, 1$ and the graph of f with the corresponding contour lines on the graph.



Further examples of level sets of a function

Example 4.1.17

This figure shows a circular disk, D, and all the points in D where the value of a function $t(x, y)$ is $2, 3, 4, 5, 6, 8, 10$ or 10.5.



- The level sets $t = 2$ and $t = 10.5$ are points.
- The leve lsets $t = 3, t = 4, t = 6$, and $t = 8$ are curves.
- The level set $t = 5$ consists of all the points of the shaded band.

Example 4.1.18

Let:

$$f(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$$

- The domain of f is the set of all points $(x,y,z) \neq (0,0,0)$.
- The graph of f is the set of points $(x,y,z,f(x,y,z))$.
- The c level set of f for $c > 0$ is the set of points that satisfy

$$\frac{1}{\sqrt{x^2+y^2+z^2}} = c$$

or $x^2+y^2+z^2 = \frac{1}{c^2}$.

It is the sphere with radius $\frac{1}{c}$ centered at the origin of $\mathbb{R}^3$.

Section 2

Continuity

Continuity

Motivation:
In single variable calculus, we motivated the definition of continuity of f at x by asking:

Is approximate knowledge of x sufficient to give approximate knowledge of $f(x)$?

Recall:
A function f is continuous at x if for every tolerance $\epsilon > 0$ for error in the output, we can find a level of precision $\delta > 0$ for the input, such that:

$$\text{if } |x - y| < \delta, \text{ then } |f(x) - f(y)| < \epsilon.$$

For a function $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$, continuity is defined similarly.

Definition 4.2.1

A function $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$ **is continuous** at $\mathbf{X}$ in D, if:

- For every tolerance $\epsilon > 0$, there exists a precision $\delta > 0$ (depending on ϵ), such that:

$$\text{if } \|\mathbf{X} - \mathbf{Y}\| < \delta, \text{ then } \|\mathbf{F}(\mathbf{X}) - \mathbf{F}(\mathbf{Y})\| < \epsilon \quad (\mathbf{Y} \text{ in } D).$$

Continuity of Linear Functions

Let $\mathbf{L} : \mathbb{R}^m \to \mathbb{R}^n$ be a linear function. We show that $\mathbf{L}$ is continuous at every $\mathbf{X}$.

- **Definition of Continuity**:
  For every tolerance $\epsilon > 0$, there exists a precision $\delta > 0$ such that:

$$\text{if } \|\mathbf{Y} - \mathbf{X}\| < \delta, \text{ then } \|\mathbf{L}(\mathbf{Y}) - \mathbf{L}(\mathbf{X})\| < \epsilon.$$

- **Variable Substitution**:
  Let $\mathbf{Y} = \mathbf{X} + \mathbf{H}$. Then:

$$\|\mathbf{L}(\mathbf{Y}) - \mathbf{L}(\mathbf{X})\| = \|\mathbf{L}(\mathbf{H})\|.$$

Continue

- **Matrix Representation**:
  By **Theorem 4.1.6**, there exists a matrix $\mathbf{C}$ such that:

$$\mathbf{L}(\mathbf{H}) = \mathbf{C}\mathbf{H}.$$

  By **Theorem 4.1.8**:

$$\|\mathbf{L}(\mathbf{H})\| = \|\mathbf{C}\mathbf{H}\| \leq \|\mathbf{C}\|\|\mathbf{H}\|.$$

- **Case 1:** $\|\mathbf{C}\| = 0$

  If $\|\mathbf{C}\| = 0$, then $\mathbf{L}$ is the constant function $\mathbf{0}$. Constant functions are continuous at each $\mathbf{X}$ (because $\|\mathbf{F}(\mathbf{X}) - \mathbf{F}(\mathbf{Y})\| = 0$).

- **Case 2:** $\|\mathbf{C}\| \neq 0$

  Given $\epsilon > 0$, take $\|\mathbf{H}\| < \delta = \frac{\epsilon}{\|\mathbf{C}\|}$. Then:

$$\|\mathbf{L}(\mathbf{Y}) - \mathbf{L}(\mathbf{X})\| = \|\mathbf{L}(\mathbf{H})\| \leq \|\mathbf{C}\|\|\mathbf{H}\| < \epsilon.$$

  Therefore, $\mathbf{L}$ is continuous at $\mathbf{X}$.

Definition 4.2.2

A function $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$ **is continuous** on D if $\mathbf{F}$ is continuous at every $\mathbf{X}$ in D.

Definition 4.2.3

We say that a sequence of points $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_k, \ldots$ in $\mathbb{R}^n$ **converges to** $\mathbf{X}$, if:

- For every $\epsilon > 0$ there is a whole number N so that if $k \geq N$, then $\|\mathbf{X}_k - \mathbf{X}\| < \epsilon$.

As with functions from $\mathbb{R}$ to $\mathbb{R}$, a continuous function takes a convergent sequence of points in the domain to a convergent sequence of points in the range.

Theorem 4.2.4

If $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$ *is continuous on* D*, then for every sequence*

$$\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_k, \ldots$$

of points in D *that converges to a point* $\mathbf{X}$ *in* D*, the sequence*

$$\mathbf{F}(\mathbf{X}_1), \mathbf{F}(\mathbf{X}_2), \ldots, \mathbf{F}(\mathbf{X}_k), \ldots$$

converges to $\mathbf{F}(\mathbf{X})$*.*

Proof

Take $\epsilon > 0$. Since $\mathbf{F}$ is continuous at $\mathbf{X}$, there is $\delta > 0$ so that

$$\text{if } \|\mathbf{X} - \mathbf{Y}\| < \delta, \text{ then } \|\mathbf{F}(\mathbf{X}) - \mathbf{F}(\mathbf{Y})\| < \epsilon$$

Since the $\mathbf{X}_k$ converge to $\mathbf{X}$, given this δ there is a whole number N so that

$$\text{if } k \geq N, \text{ then } \|\mathbf{X} - \mathbf{X}_k\| < \delta$$

Therefore

$$\text{If } k \geq N, \text{ then } \|\mathbf{F}(\mathbf{X}) - \mathbf{F}(\mathbf{X}_k)\| < \epsilon$$

This proves that the sequence $\mathbf{F}(\mathbf{X}_k)$ converges to $\mathbf{F}(\mathbf{X})$.

The continuity problem of $\mathbf{F}$ can be simplified to check the continuity of each component function.

Theorem 4.2.5

A function $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$, *denoted*

$$\mathbf{F}(\mathbf{X}) = (f_1(\mathbf{X}), f_2(\mathbf{X}), \ldots, f_m(\mathbf{X}))$$

is continuous on D *if and only if each component function* $f_j : D \subset \mathbb{R}^n \to \mathbb{R}$ *is continuous on* D.

Proof

Necessity:
If $\mathbf{F}$ is continuous at $\mathbf{X}$, then each component function f_i is continuous at $\mathbf{X}$.

- Suppose $\mathbf{F}$ is continuous at $\mathbf{X}$.
- For every $\epsilon > 0$, there exists $\delta > 0$ such that if $\|\mathbf{Y} - \mathbf{X}\| < \delta$, then $\|\mathbf{F}(\mathbf{Y}) - \mathbf{F}(\mathbf{X})\| < \epsilon$.
- Let f_i be one of the component functions. Since each component of a vector has absolute value less than or equal to the norm of the vector, we have:

$$|f_i(\mathbf{Y}) - f_i(\mathbf{X})| \leq \|\mathbf{F}(\mathbf{Y}) - \mathbf{F}(\mathbf{X})\| < \epsilon.$$

Thus, continuity of $\mathbf{F}$ at $\mathbf{X}$ implies continuity of each component function f_i at $\mathbf{X}$.

Proof (Continued)

Sufficiency:
If each component function f_i is continuous at $\mathbf{X}$, then $\mathbf{F}$ is continuous at $\mathbf{X}$.

- Suppose each f_i is continuous at $\mathbf{X}$.
- For every $\epsilon > 0$, there exist $\delta_1, \delta_2, \ldots, \delta_m$ such that if $\|\mathbf{Y} - \mathbf{X}\| < \delta_i$, then $|f_i(\mathbf{Y}) - f_i(\mathbf{X})| < \epsilon$.
- Let $\delta = \min\{\delta_1, \delta_2, \ldots, \delta_m\}$.
- If $\|\mathbf{Y} - \mathbf{X}\| < \delta$, then for all $i = 1, 2, \ldots, m$:

$$|f_i(\mathbf{Y}) - f_i(\mathbf{X})| < \epsilon.$$

Therefore:

$$\|\mathbf{F}(\mathbf{Y}) - \mathbf{F}(\mathbf{X})\| = \sqrt{\sum_{i=1}^{m} (f_i(\mathbf{Y}) - f_i(\mathbf{X}))^2} \leq \sqrt{m}\epsilon.$$

Since ϵ is arbitrary, $\mathbf{F}$ is continuous at $\mathbf{X}$.
(Since $\|\mathbf{F}(\mathbf{Y}) - \mathbf{F}(\mathbf{X})\|$ can be made as small as we like by taking δ small enough)

A very useful principle

Just now we showed that:

for every $\epsilon > 0$, we can find $\delta > 0$ such that if $\|\mathbf{Y} - \mathbf{X}\| < \delta$, then

$$\|\mathbf{F}(\mathbf{Y}) - \mathbf{F}(\mathbf{X})\| \leq C\epsilon,$$

where $C > 0$ is a constant.

Because of the constant C, the above condition seems a bit different from what we have in the definition of continuity. We claim that the above condition already implies continuity of $\mathbf{F}$ at $\mathbf{X}$.

The reason is:

the above claim is true for arbitrary $\epsilon > 0$!

Given $\epsilon > 0$, we can set $\epsilon' := \epsilon/C$. For this $\epsilon' > 0$, we can find $\delta > 0$ such that if $\|\mathbf{Y} - \mathbf{X}\| < \delta$, then

$$\|\mathbf{F}(\mathbf{Y}) - \mathbf{F}(\mathbf{X})\| \leq C\epsilon' = C\frac{\epsilon}{C} = \epsilon,$$

showing continuity of $\mathbf{F}$ at $\mathbf{X}$.

Proof (Continued)

Special Case:

Taking $n = 1$ in the **Theorem 4.2.5** shows that $\mathbf{F} : D \subset \mathbb{R}^m \to \mathbb{R}$ defined by:

$$\mathbf{F}(t) = (f_1(t), f_2(t), \ldots, f_m(t))$$

is continuous on an interval if and only if each component function f_i is continuous there.

Example 4.2.6

$\mathbf{F}(t) = (\cos t, \sin t, t)$ is continuous at t because each component function is continuous at t.

The next two theorems help us find continuous functions from $\mathbb{R}^n$ to $\mathbb{R}$.

Theorem 4.2.7

If : $D \subset \mathbb{R}^n \to \mathbb{R}$ *and* $g : D \subset \mathbb{R}^n \to \mathbb{R}$ *are continuous on* D *then:*

(a) $f + g$ *is continuous on* D,

(b) fg *is continuous on* D,

(c) $\frac{1}{g}$ *is continuous at every point* $\mathbf{X}$ *where* $g(\mathbf{X}) \neq 0$.

Proof

Proof of (a):

Let $\mathbf{X}$ be in D and $\epsilon > 0$. By continuity of f and g, there exist two precisions, δ_f and δ_g such that:

$$\|\mathbf{X} - \mathbf{Y}\| < \delta_f \Rightarrow |f(\mathbf{X}) - f(\mathbf{Y})| < \epsilon, \tag{4.4}$$

$$\|\mathbf{X} - \mathbf{Y}\| < \delta_g \Rightarrow |g(\mathbf{X}) - g(\mathbf{Y})| < \epsilon. \tag{4.5}$$

Let $\delta = \min\{\delta_f, \delta_g\}$. If $\|\mathbf{X} - \mathbf{Y}\| < \delta$, then according to the triangle inequality:

$$|f(\mathbf{X}) - f(\mathbf{Y}) + g(\mathbf{X}) - g(\mathbf{Y})| \leq |f(\mathbf{X}) - f(\mathbf{Y})| + |g(\mathbf{X}) - g(\mathbf{Y})| < 2\epsilon$$

$$\Downarrow$$

$$|(f+g)(\mathbf{X}) - (f+g)(\mathbf{Y})| \leq |f(\mathbf{X}) - f(\mathbf{Y})| + |g(\mathbf{X}) - g(\mathbf{Y})| < 2\epsilon.$$

Since 2ϵ can be made arbitrarily small, $f + g$ is continuous.

Proof (Continued)

Proof of (b):

Let ϵ be a tolerance and let δ be as in the proof of part (a).

By algebra and the triangle inequality:

$$|f(\mathbf{X})g(\mathbf{X}) - f(\mathbf{Y})g(\mathbf{Y})| \leq |f(\mathbf{X})||g(\mathbf{X}) - g(\mathbf{Y})| + |g(\mathbf{Y})||f(\mathbf{X}) - f(\mathbf{Y})|.$$

If $\|\mathbf{X} - \mathbf{Y}\| < \delta$, then we have inequalities (4.4) and (4.5), therefore:

$$|f(\mathbf{X})g(\mathbf{X}) - f(\mathbf{Y})g(\mathbf{Y})| < \epsilon(|f(\mathbf{X})| + |g(\mathbf{Y})|).$$

We also know by the triangle inequality that

$$\Big||g(\mathbf{X})| - |g(\mathbf{Y})|\Big| \leq |g(\mathbf{X}) - g(\mathbf{Y})| < \epsilon$$

so $|g(\mathbf{Y})| < |g(\mathbf{X})| + \epsilon$. Therefore

$$|f(\mathbf{X})g(\mathbf{X}) - f(\mathbf{Y})g(\mathbf{Y})| < \epsilon(|f(\mathbf{X})| + |g(\mathbf{X})| + \epsilon).$$

For a given $\mathbf{X}$, $f(\mathbf{X})$ and $g(\mathbf{X})$ are fixed numbers, so the expression on the right side can be made arbitrarily small, which implies that fg is continuous.

Proof (Continued)

Proof of (c):

Let $g(\mathbf{X}) = k \neq 0$. By continuity of g, there exists $\gamma > 0$ such that:

$$\|\mathbf{X} - \mathbf{Y}\| < \gamma \Rightarrow |g(\mathbf{Y}) - g(\mathbf{X})| < \left|\frac{1}{2}k\right| \Rightarrow |g(\mathbf{Y})| > \left|\frac{1}{2}k\right|.$$

For $\epsilon > 0$, there exists $\delta > 0$ such that:

$$\|\mathbf{X} - \mathbf{Y}\| < \delta \Rightarrow |g(\mathbf{X}) - g(\mathbf{Y})| < \epsilon.$$

If $\|\mathbf{X} - \mathbf{Y}\| < \min\{\gamma, \delta\}$, then:

$$\left|\frac{1}{g(\mathbf{X})} - \frac{1}{g(\mathbf{Y})}\right| = \frac{|g(\mathbf{X}) - g(\mathbf{Y})|}{|g(\mathbf{X})||g(\mathbf{Y})|} < \frac{\epsilon}{\left|\frac{1}{2}k^2\right|}.$$

Since k is fixed and ϵ can be made arbitrarily small, $\frac{1}{g}$ is continuous.

Example 4.2.8

Consider the function from $\mathbb{R}^3$ to $\mathbb{R}$,

$$f(x, y, z) = \frac{x^2 + xy - 2x + z + 7}{x^2 - y^3}$$

- The **numerator** $x^2 + xy - 2x + z + 7$ is continuous at every (x, y, z) because it is a sum of:
  - Constants (e.g., 7),
  - Linear functions (e.g., z),
  - Products of constants and linear functions (e.g., xy).
- The **denominator** $x^2 - y^3$ is also continuous for the same reason.
- According to **Theorem 4.2.7**, f is continuous at every point (x, y, z) where the denominator $x^2 - y^3 \neq 0$.

Theorem 4.2.9

If $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$ is continuous on D, $g : A \subset \mathbb{R}^m \to \mathbb{R}$ is continuous on A, and the range of $\mathbf{F}$ is contained in A, then the composition $g \circ \mathbf{F} : D \to \mathbb{R}$ is continuous on D.

Proof: Let $\mathbf{X}$ be in D.

- Since g is continuous at $\mathbf{F}(\mathbf{X})$, for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$\|\mathbf{F}(\mathbf{X}) - \mathbf{Y}\| < \delta \Rightarrow |g(\mathbf{F}(\mathbf{X})) - g(\mathbf{Y})| < \epsilon.$$

- Since $\mathbf{F}$ is continuous at $\mathbf{X}$, there exists $\gamma > 0$ such that:

$$\|\mathbf{X} - \mathbf{Z}\| < \gamma \Rightarrow \|\mathbf{F}(\mathbf{X}) - \mathbf{F}(\mathbf{Z})\| < \delta.$$

- Combining these, for $\|\mathbf{X} - \mathbf{Z}\| < \gamma$, we have:

$$|g(\mathbf{F}(\mathbf{X})) - g(\mathbf{F}(\mathbf{Z}))| < \epsilon.$$

- Therefore, $g \circ \mathbf{F}$ is continuous at $\mathbf{X}$.

Theorem 4.2.10

If $\mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^m$ *is continuous on* $D, \mathbf{G} : A \subset \mathbb{R}^m \to \mathbb{R}^k$ *is continuous on* A *and the range of* $\mathbf{F}$ *is contained in* A*, then the composition* $\mathbf{G} \circ \mathbf{F} : D \subset \mathbb{R}^n \to \mathbb{R}^k$ *is continuous on* D*.*

Proof:

Denote $\mathbf{G}(y_1, \ldots, y_m) = (g_1(y_1, \ldots, y_m), \ldots, g_k(y_1, \ldots, y_m))$.

- By **Theorem 4.2.5**, each component function g_i is continuous on A.
- By **Theorem 4.2.9**, each composition $g_i \circ \mathbf{F}$ is continuous on D.
- By **Theorem 4.2.5**, a vector-valued function whose components are continuous is continuous.
- Therefore, $\mathbf{G} \circ \mathbf{F}$ is continuous on D.

Example 4.2.11

Let $\mathbf{F}$ be the function

$$\mathbf{F}(\mathbf{X}) = -\frac{\mathbf{X}}{\|\mathbf{X}\|^3}, \quad \mathbf{X} \neq \mathbf{0}$$

Rewriting $\mathbf{F}$ in coordinate notation,

$$\mathbf{F}(x, y, z) = \left(\frac{-x}{\left(x^2 + y^2 + z^2\right)^{3/2}}, \frac{-y}{\left(x^2 + y^2 + z^2\right)^{3/2}}, \frac{-z}{\left(x^2 + y^2 + z^2\right)^{3/2}} \right).$$

- By **Theorem 4.2.9**, the function $\left(x^2 + y^2 + z^2\right)^{3/2}$ is continuous.
- By **Theorem 4.2.7**, each component

$$f_1(x, y, z) = \frac{-x}{\left(x^2 + y^2 + z^2\right)^{3/2}}$$

  is a continuous function at every (x, y, z) in $\mathbb{R}^3$ except the origin, so f_1 is continuous on $\mathbb{R}^3 - \{(0, 0, 0)\}$. Similarly the components f_2 and f_3 are continuous on $\mathbb{R}^3 - \{(0, 0, 0)\}$.
- So by **Theorem 4.2.5**, $\mathbf{F}$ is continuous on $\mathbb{R}^3 - \{(0, 0, 0)\}$.

Example 4.2.12

Consider

$$\mathbf{F}(x,y,z) = \left(\sin(x+y), \mathrm{e}^{xz+y^2}, \log(xz)\right)$$

- The component functions:

  $$f_1(x,y,z) = \sin(x+y), \quad f_2(x,y,z) = \mathrm{e}^{xz+y^2}$$

  are continuous at every point (x,y,z).

  $$f_3(x,y,z) = \log(xz)$$

  is continuous at each point (x,y,z) where $xz > 0$.
- By **Theorem 4.2.5**, the vector field $\mathbf{F}$ is continuous at each point (x,y,z) where $xz > 0$.
- Alternatively, define the functions:

  $$\mathbf{H}(x,y,z) = \left(x+y, xz+y^2, xz\right), \quad \mathbf{G}(u,v,w) = (\sin u, \mathrm{e}^v, \log w)$$

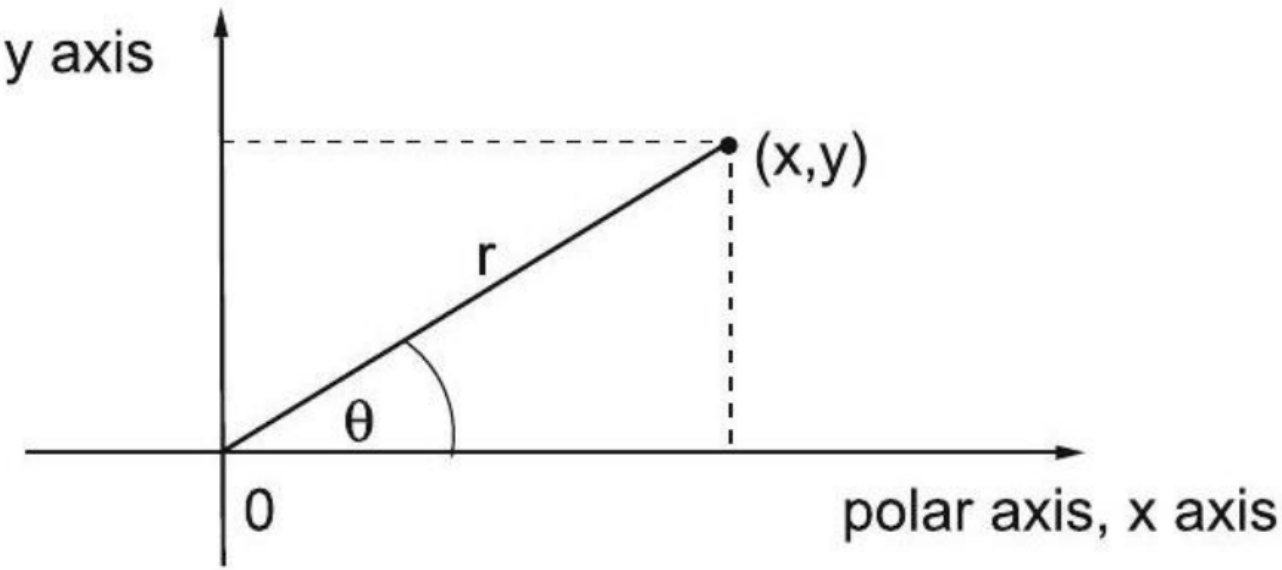
  Then, $\mathbf{G} \circ \mathbf{H} = \mathbf{F}$.
- By **Theorem 4.2.10**, $\mathbf{F}$ is continuous at all points (x,y,z) where $xz > 0$.

Section 3

Other coordinate systems

Polar Coordinates

It is often convenient to use polar coordinates (r, θ) rather than rectangular coordinates (x, y) to describe a curve or a region in the plane.



This figure shows the coordinates of a point in a plane given in terms of rectangular coordinates and polar coordinates, where we usually take

$$r \geq 0 \quad \text{and} \quad 0 \leq \theta \leq 2\pi$$

Polar coordinates

The coordinates are related by

$$x = r\cos\theta$$
$$y = r\sin\theta \tag{4.6}$$

and

$$r = \sqrt{x^2 + y^2} \tag{4.7}$$

Let $\mathbf{F} : \mathbb{R}^2 \to \mathbb{R}^2$ be the function defined by

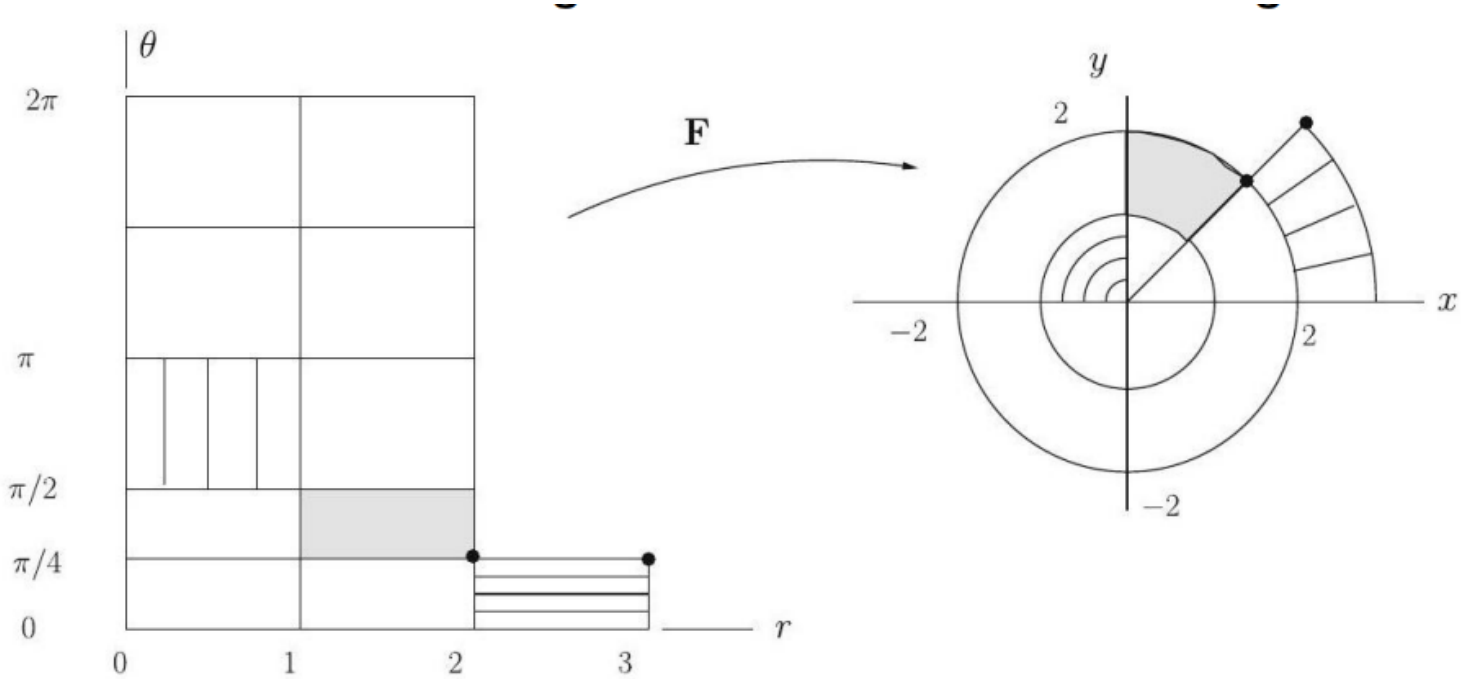
$$\mathbf{F}(r, \theta) = (r\cos\theta, r\sin\theta)$$

Polar coordinates

The component functions of $\mathbf{F}$ are: $x = r\cos\theta, \quad y = r\sin\theta.$

$\mathbf{F}$ is called the polar coordinate mapping. Let's see how regions in the (r, θ) plane are mapped to the (x, y) plane. See in the following figure.

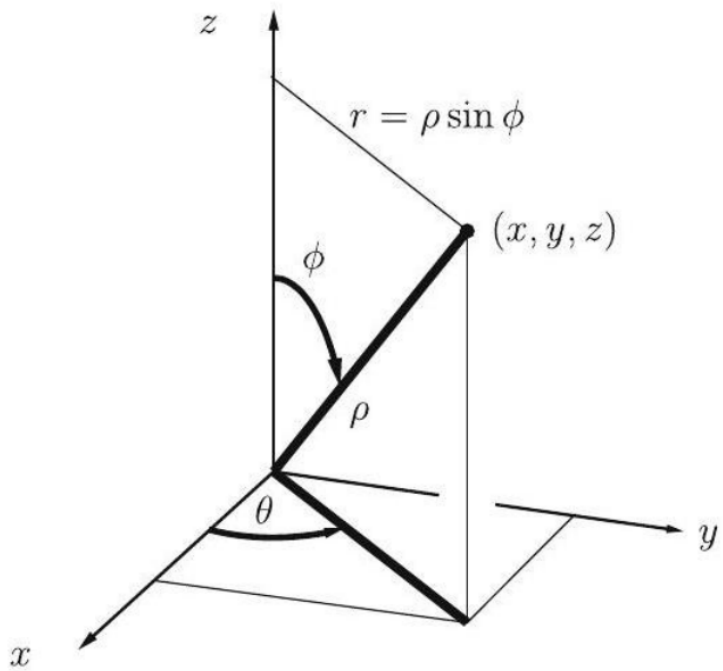
- If we restrict $\mathbf{F}$ to a rectangular region $0 < a \le r \le b$, $0 < \alpha \le \theta \le \beta < 2\pi$, then $\mathbf{F}$ is one to one.
- Three such regions are indicated in the figure.



Spherical coordinates

Another way to describe the location of a point (x, y, z) is:

- Denote by ρ the distance between the point and the origin.
- Let ϕ be the angle between the positive z axis and the line through $(0, 0, 0)$ and (x, y, z).
- Let θ be the angle between the plane containing (x, y, z) and the z axis, and the x, z plane.



Definition 4.3.1

The spherical coordinates (ρ, ϕ, θ) that correspond to the Cartesian coordinates (x, y, z) are related by:

$$x = \rho \sin\phi \cos\theta$$
$$y = \rho \sin\phi \sin\theta$$
$$z = \rho \cos\phi$$

$0 \leq \rho,\ 0 \leq \phi \leq \pi,\ 0 \leq \theta \leq 2\pi.$

Example 4.3.2

The set of points in space that lie on a sphere of radius 3 centered at the origin satisfy

$$\sqrt{x^2 + y^2 + z^2} = 3$$

Using **spherical coordinates** we describe these points by

$$\rho = 3$$

and ϕ and θ take their full range: $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi$.

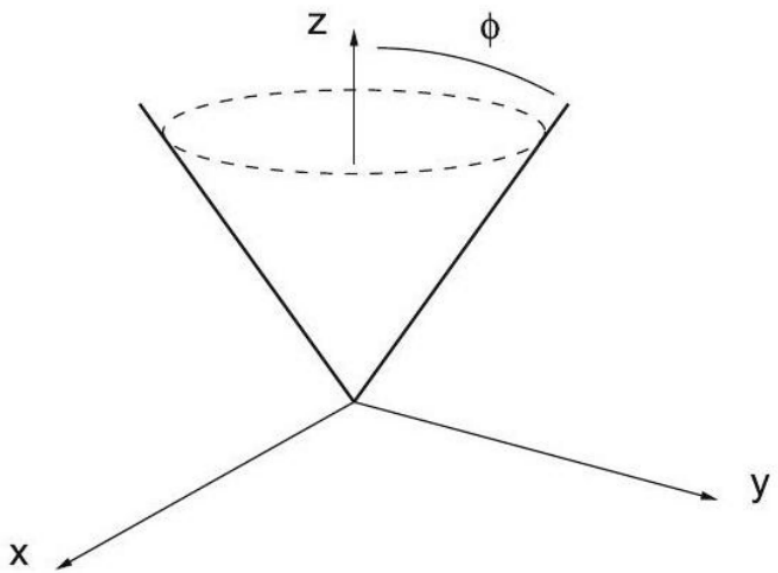
Circular cone surface

Example 4.3.3

To describe the set of points on the circular cone surface

$$z = \sqrt{x^2 + y^2}$$

shown in the figure we note that every point on the cone has the same angle ϕ.



To find ϕ we look at the intersection of the cone with the plane $y = 0$.

At such points we get $z = \sqrt{x^2} = |x|$, so $\phi = \frac{\pi}{4}$, and ρ and θ are unrestricted.

Continue

Another approach is to substitute the formulas:

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho \cos\phi$$

into the equation for the cone:

$$\begin{aligned} \rho \cos\phi &= \sqrt{\rho^2 \sin^2\phi \cos^2\theta + \rho^2 \sin^2\phi \sin^2\theta} \\ &= \rho \sin\phi. \end{aligned}$$

Since $0 \leq \phi \leq \pi$, this is satisfied only by $\phi = \frac{\pi}{4}$.

Example 4.3.4

Let D be the solid region of points that satisfy

$$x^2 + y^2 + z^2 \leq 9, \quad x^2 + y^2 + z^2 > 1, \quad z \geq 0$$

The surface of the inner hemisphere is sketched with dots to indicate that the points on that surface are not included.

Region D is described in spherical coordinates as

$$1 < \rho \leq 3, \quad 0 \leq \phi \leq \frac{\pi}{2}$$

